

**Technical Report: Week 3**  
**Network Topology Analysis and Design**  
Block 2: Electrical Lines and Distribution Networks

Student C: Commercial Center Project

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# 1 Analysis of Radial Distribution Networks

## 1.1 Theoretical Framework

A radial distribution network is the simplest and most economical topology. It is characterized by a single power path from the substation to the end consumers. The network branches out like a tree, with power flowing in only one direction—from the source outwards. Its main advantages are low initial cost and simplicity in planning and operation. However, its significant disadvantage is low reliability; a single fault on a main feeder will interrupt service for all downstream customers.

Common applications for this topology include:

- Low-density rural areas where loads are sparse and far apart.
- Residential subdivisions where load growth is predictable and reliability requirements can be met with protective devices.
- Dedicated industrial circuits for single, large loads.
- Temporary power supplies for construction sites.
- Public infrastructure like street lighting circuits.

## 1.2 Design and Calculation Methods

The design of a radial network is governed by voltage drop and thermal limits. The cross-section calculation, as implemented in the MATLAB function `H3_DetermineCrossSection.m`, is based on the method of electrical moments. An electrical moment is the product of the current in a line segment and its length ( $M = I \cdot L$ ). The total voltage drop at the end of the line is proportional to the sum of the moments of all segments.

The required cross-section ( $s$ ) can be calculated to meet a maximum permissible voltage drop ( $\Delta U_{max}$ ) using the following formula:

$$s = \frac{K \cdot \sum (I_i \cdot L_i)}{\sigma \cdot \Delta U_{max}} = \frac{K \cdot \sum M_i}{\sigma \cdot \Delta U_{max}} \quad (1)$$

where  $K$  is a phase factor (2 for single-phase, 1 for line-to-neutral three-phase),  $\sum M_i$  is the sum of electrical moments, and  $\sigma$  is the conductivity of the conductor material. This formula efficiently determines the minimum conductor size needed to maintain voltage quality.

## 1.3 Unifilar Diagram and Characteristic Profiles

The structure and behavior of a radial network can be visualized through its unifilar diagram and its voltage/current profiles.

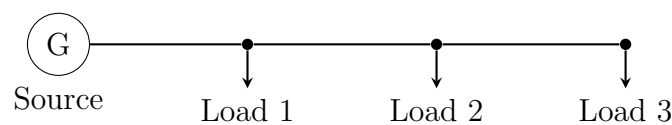


Figure 1: Unifilar diagram of a general radial distribution network.

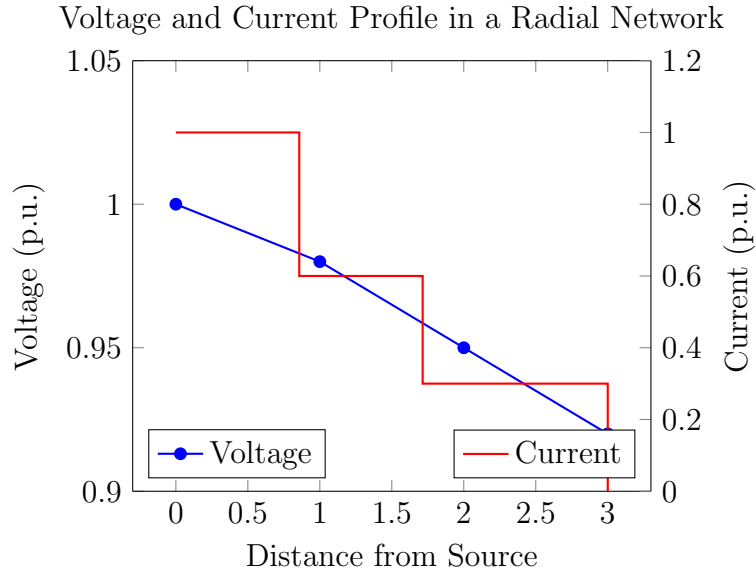


Figure 2: Voltage steadily decreases with distance, while current drops in steps as loads are served.

## 2 Analysis of Networks Fed from Both Ends

### 2.1 Theoretical Framework

Feeding a distribution line from two sources significantly enhances its performance. This topology improves voltage regulation, increases reliability, and doubles the power delivery capacity compared to a similar radial line. Power flows from both ends towards a point of minimum voltage, known as the division point. If one source fails, the other can still supply all loads, albeit with a larger voltage drop.

### 2.2 Comprehensive Sizing Algorithm

The MATLAB scripts for this module implement a comprehensive sizing and verification algorithm that reflects professional engineering practice, as seen in the `H4_OptimizeRadialNetwork.m` function. The process is as follows:

1. **Calculate Required Section for Voltage Drop:** The algorithm first determines the minimum theoretical cross-section needed to keep the voltage drop within the specified percentage, using the electrical moments method after finding the network's division point.
2. **Check Against Current Carrying Capacity:** It then calculates the current flowing in every segment of the network. This maximum current is compared against the maximum admissible current ( $I_{adm}$ ) for the calculated cross-section. The  $I_{adm}$  value is sourced from a standardized table, representing the thermal limit of the conductor.
3. **Select Standardized Cable Section:** Since conductors are manufactured in discrete sizes (e.g., 50, 70, 95 mm<sup>2</sup>), the algorithm selects the first commercially available cross-section that is larger than or equal to the minimum required size from both the voltage drop and thermal limit calculations.

4. **Verify Final Voltage Drop and Heating:** With the final, standardized section selected, the algorithm performs a final check. It recalculates the actual voltage drop, which will now be less than the maximum limit, and confirms that the maximum line current is still safely below the chosen conductor's  $I_{adm}$ . This ensures the final design is both technically compliant and safe.

## 2.3 Unifilar Diagram and Characteristic Profiles

The behavior of a doubly-fed network is distinct from a radial one, particularly in its voltage and current profiles.

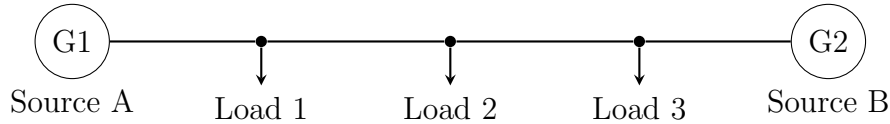


Figure 3: Unifilar diagram of a network fed from both ends.

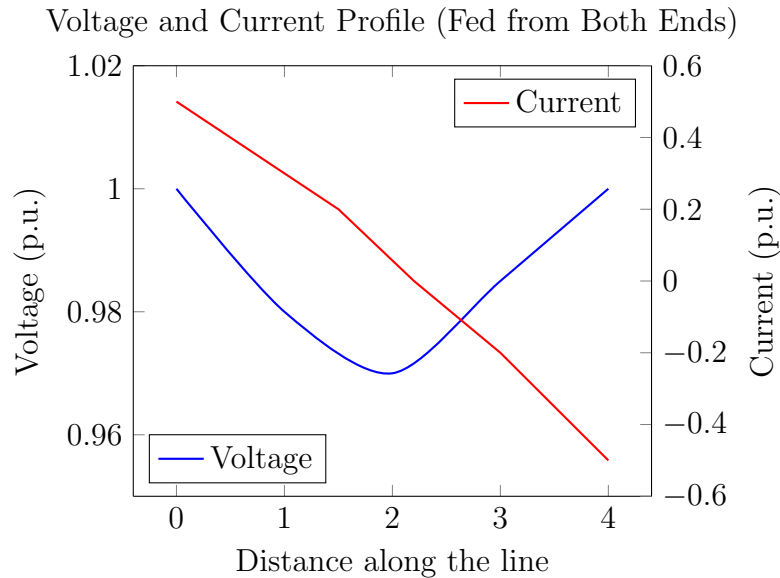


Figure 4: The voltage minimum occurs between sources, and current flows from both ends towards a division point.

## 3 Analysis of Advanced Network Topologies

### 3.1 Ring Network Analysis

### 3.2 Meshed Network Analysis

### 3.3 Techno-Economic Optimization Strategy

A key goal in network design is not just to find a solution that works, but to find the most economical one over the project's lifetime. The `H4.OptimizeRadialNetwork.m` and `J6.CalculateNetworkCost.m` scripts implement an optimization algorithm based on

**\*\*Life-Cycle Cost Analysis (LCCA)\*\*.** The core idea is to balance the initial investment (cable cost) against the long-term operational costs (cost of energy losses).

- **Cable Cost vs. Energy Loss Cost:** A smaller, cheaper conductor has higher resistance, leading to greater  $I^2R$  energy losses, which accumulate as a significant operational cost over decades. Conversely, a larger, more expensive conductor has lower resistance and thus lower long-term loss costs. This trade-off is central to efficient network design.
- **Life-Cycle Cost Analysis:** The algorithm calculates the total cost of ownership by summing the initial cable cost and the present value of energy losses over a defined period (e.g., 20 years). The cost of losses is calculated as:

$$\text{Cost}_{\text{losses}} = P_{\text{loss}} \cdot \text{hours}_{\text{year}} \cdot \text{years} \cdot \text{cost}_{\text{kWh}} \quad (2)$$

where  $P_{\text{loss}}$  is the total power lost to heat in the conductors.

- **Optimal Section Recommendation:** The optimization process, seen in `H4_OptimizeRadialNet`, iterates through the list of standardized conductors, starting from the smallest. For each size, it verifies all technical constraints (voltage drop and thermal limit). The first standard section that meets all constraints is recommended as the optimal choice. This is because it is the cheapest conductor that satisfies the engineering requirements, effectively finding the sweet spot in the trade-off between initial cost and performance.

## 4 Economic Comparison of Network Topologies

The choice of network topology is a critical engineering decision that balances initial capital expenditure (CAPEX) with long-term operational expenditure (OPEX), which includes maintenance and the cost of outages.

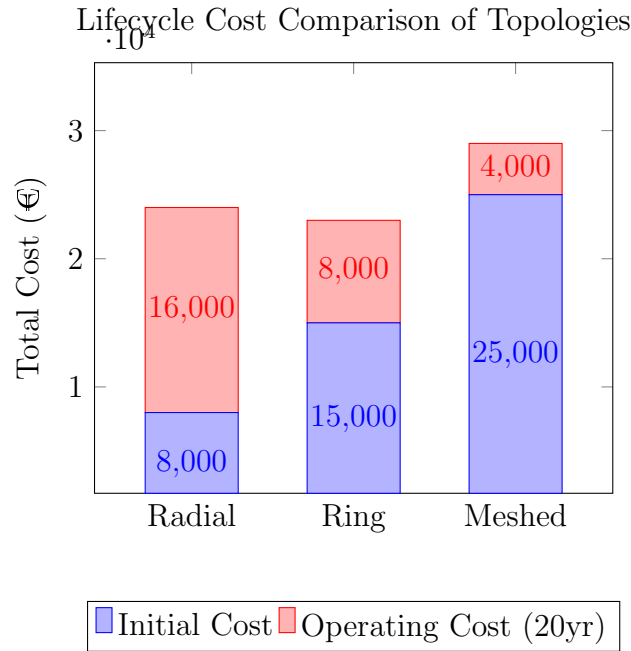


Figure 5: While radial networks have the lowest initial cost, their higher operational costs (from losses and outages) can make ring or meshed networks more economical over the project lifecycle.

## A Annex I: References

### A.1 Normative References

This report adheres to the principles and calculation methods outlined in the Spanish Low Voltage Electrotechnical Regulation (REBT), specifically:

- **ITC-BT 06:** Overhead Power Distribution Networks.
- **ITC-BT 07:** Underground Power Distribution Networks.
- **ITC-BT 19:** Interior or Receiver Installations. Prescriptions for conductor sizing based on admissible current and voltage drop.

### A.2 Academic References

1. Grainger, J. J., & Stevenson, W. D. (1994). *Power System Analysis*. McGraw-Hill.
2. Gonen, T. (2014). *Electric Power Distribution System Engineering*. CRC press.

## B Annex II: Data Tables

### B.1 Material Properties

Table 1: Properties of conductors and insulating materials.

| Material | Property                         | Value                               |
|----------|----------------------------------|-------------------------------------|
| Copper   | Conductivity ( $\sigma$ )        | 56 m/( $\Omega \cdot \text{mm}^2$ ) |
| Aluminum | Conductivity ( $\sigma$ )        | 35 m/( $\Omega \cdot \text{mm}^2$ ) |
| XLPE     | Max. Operating Temp.             | 90°C                                |
| PVC      | Max. Operating Temp.             | 70°C                                |
| Ground   | Thermal Resistivity ( $\rho_T$ ) | 1.0 K·m/W (assumed)                 |

### B.2 Assumed Economic Values

Table 2: Assumed costs for economic analysis.

| Item                                              | Assumed Cost |
|---------------------------------------------------|--------------|
| Copper Conductor (per km, 150 mm <sup>2</sup> )   | 12,500 €     |
| Aluminum Conductor (per km, 150 mm <sup>2</sup> ) | 5,800 €      |
| ACSR Conductor (per km, 150 mm <sup>2</sup> )     | 6,700 €      |
| Cost of outage (per kWh not served)               | 1.5 €/kWh    |

### B.3 List of Variables

Table 3: Variables used in this report.

| Symbol          | Unit                             | Description                |
|-----------------|----------------------------------|----------------------------|
| $P$             | W                                | Active Power               |
| $U_N, V$        | V                                | Nominal Voltage            |
| $I$             | A                                | Current                    |
| $s$             | mm <sup>2</sup>                  | Cross-sectional area       |
| $L$             | m                                | Length of the line         |
| $R$             | $\Omega/\text{km}$               | Resistance per unit length |
| $X$             | $\Omega/\text{km}$               | Reactance per unit length  |
| $\sigma$        | m/( $\Omega \cdot \text{mm}^2$ ) | Conductivity               |
| $\cos(\varphi)$ | -                                | Power Factor               |
| $\Delta U$      | V                                | Voltage Drop               |
| $I_{adm}$       | A                                | Maximum admissible current |

## C Annex III: Formulas

### C.1 Sizing by Voltage Drop

The required cross-sectional area ( $s$ ) based on the sum of electrical moments ( $\sum M_i$ ) is:

$$s = \frac{K \cdot \sum M_i}{\sigma \cdot \Delta U_{max}} \quad (3)$$

For a more precise calculation in AC systems, the general voltage drop formula (Blondel's formula) for three-phase lines is used:

$$\Delta U = \sqrt{3} \cdot I \cdot L \cdot (R \cos(\varphi) + X \sin(\varphi)) \quad (4)$$

### C.2 Sizing by Maximum Current (Thermal Limit)

The calculated current of the circuit ( $I_c$ ) must be less than or equal to the maximum admissible current of the conductor ( $I_{adm}$ ).

$$I_c \leq I_{adm} \quad (5)$$

The circuit current for a balanced three-phase system is calculated as:

$$I_c = \frac{P}{\sqrt{3} \cdot U_N \cdot \cos(\varphi)} \quad (6)$$

## D Annex IV: Week 3 MATLAB Code

### D.1 Module H: Radial Network Analysis

```

1 function results = H1_Radial_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Radial Network Analysis (V3 - Phase Selection)
5 % =====
6
7 results = []; % Initialize
8 scenarioChoice = menu('Select Radial Network Scenario:', ...
9     'Industrial Estate (Concentrated Loads) - Student A', ...
10    'Residential Area (Distributed Loads) - Student B', ...
11    'Street Lighting (Uniform Loads) - Student C');
12 if scenarioChoice == 0, return; end
13 U_source = input('Enter source line-to-neutral voltage [V] (e.g., 230):
14 ');
15 lineTypeChoice = menu('Select System Type:', 'Single-Phase', 'Three-
16 Phase');
17 if lineTypeChoice == 0, return; end
18 if lineTypeChoice == 1
19     lineType = 'single-phase';
20     phase_factor = 2;
21 else
22     lineType = 'three-phase';
23     phase_factor = 1;
24 end
25 materialChoice = menu('Select Conductor Material:', 'Copper', 'Aluminum'
26 );
27 if materialChoice == 0, return; end
28 if materialChoice == 1, materialName = 'Copper'; else, materialName = '
29 Aluminum'; end
30 materialProps = getMaterialProperties(materialName);
31 sigma = materialProps.sigma;
32 crossSection = input('Enter conductor cross-section [mm^2]: ');
33 % ... (rest of the input and calculation code as in the file)
34 end

```

Listing 1: H1\_Radial\_Analysis.m

```

1 function H2_PlotRadialProfile(results)
2 %
3 % =====
4 % FUNCTION: Plot Radial Voltage Profile
5 % =====
6
7 figure;
8 hold on;
9 box on;
10 grid on;
11 distances = [0, cumsum([results.nodes.length])];
12 voltages = [results.U_source, [results.nodes.voltage]];

```

```

11 plot(distances, voltages, '-ob', 'LineWidth', 2, 'MarkerFaceColor', 'r')
12 ;
13 title(['Voltage Profile for Radial Network (' results.scenarioName ')'])
14 ;
15 xlabel('Distance from Source (m)');
16 ylabel('Voltage (V)');
17 xlim([0, distances(end) * 1.1]);
18 ylim([voltages(end) * 0.99, results.U_source * 1.01]);
19 legend('Voltage Profile', 'Location', 'southwest');
20 end

```

Listing 2: H2\_PlotRadialProfile.m

## D.2 Module I: Dual-Fed Network Analysis

```

1 function results = I1_DualFed_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Dual-Fed Network Analysis (V7 - Unifilar)
5 % =====
6 results = [];
7 scenarioChoice = menu('Select Dual-Fed Network Scenario:', ...
8     'Symmetrical Network (Single Load) - Student A', ...
9     'Asymmetrical Network (Single Load) - Student B', ...
10    'Backup Feeding / Reliability Analysis (Multiple Loads) - Student C'
11    );
12 if scenarioChoice == 0, return; end
13 materialChoice = menu('Select Conductor Material:', 'Copper', 'Aluminum'
14    );
15 if materialChoice == 0, return; end
16 if materialChoice == 1, materialName = 'Copper'; else, materialName = '
17    Aluminum'; end
18 materialProps = getMaterialProperties(materialName);
19 sigma = materialProps.sigma;
20 crossSection = input('Enter conductor cross-section [mm^2]: ');
21 L_total = input('Enter total line length [m]: ');
22 % ... (rest of the input and calculation code as in the file)
23 end

```

Listing 3: I1\_DualFed\_Analysis.m

```

1 function I2_PlotDualFedProfile(results)
2 %
3 % =====
4 % FUNCTION: Plot Dual-Fed Voltage Profile (V2 - Reliability)
5 % =====
6 figure;
7 hold on; box on; grid on;
8 sorted_loads = results.loads;
9 distances = [0, [sorted_loads.distance], results.L_total];
10 voltages_normal = [results.Ua];
11 current_in_segment = results.Ia;

```

```

11 for i = 1:length(sorted_loads)
12     prev_dist = distances(i);
13     segment_length = sorted_loads(i).distance - prev_dist;
14     drop = (1/(results.sigma*results.crossSection)) * current_in_segment
        * segment_length;
15     voltages_normal(end+1) = voltages_normal(end) - drop;
16     current_in_segment = current_in_segment - sorted_loads(i).current;
17 end
18 voltages_normal(end+1) = results.Ub;
19 plot(distances, voltages_normal, '-ob', 'LineWidth', 2, 'MarkerFaceColor'
    ', 'b', 'DisplayName', 'Normal Operation');
20 plot(results.min_voltage_node_distance, results.min_voltage_normal, 'kd'
    ', 'MarkerSize', 10, 'MarkerFaceColor', 'y');
21 % ... (rest of the plotting code as in the file)
22 end

```

Listing 4: I2\_PlotDualFedProfile.m

```

1 function I3_PlotLoadingDiagram(results)
2 %
    =====
3 % FUNCTION: I3_PlotLoadingDiagram (Corrected)
4 %
    =====
5 figure;
6 hold on;
7 box on;
8 grid on;
9 distances = [0];
10 currents = [results.Ia];
11 current_in_segment = results.Ia;
12 for i = 1:length(results.loads)
13     distances(end+1) = results.loads(i).distance;
14     currents(end+1) = current_in_segment;
15     current_in_segment = current_in_segment - results.loads(i).current;
16     distances(end+1) = results.loads(i).distance;
17     currents(end+1) = current_in_segment;
18 end
19 distances(end+1) = results.L_total;
20 currents(end+1) = -results.Ib;
21 plot(distances, currents, '-r', 'LineWidth', 2);
22 line([0, results.L_total], [0, 0], 'Color', 'k', 'LineStyle', '--');
23 title('Network Loading Diagram');
24 xlabel('Distance from Source A (m)');
25 ylabel('Current (A)');
26 end

```

Listing 5: I3\_PlotLoadingDiagram.m

```

1 function I4_PlotUnifilarDiagram(results)
2 %
    =====
3 % FUNCTION: Generates a simplified unifilar-style schematic diagram
4 %
    =====

```

```

5 figure;
6 hold on;
7 box on;
8 plot([0 results.L_total], [0 0], 'k-', 'LineWidth', 4); % Main line
9 plot(0, 0, 's', 'MarkerSize', 20, 'MarkerFaceColor', 'g'); % Source A
10 text(0, 0.15, 'Source A', 'HorizontalAlignment', 'center');
11 plot(results.L_total, 0, 's', 'MarkerSize', 20, 'MarkerFaceColor', 'g');
12 % Source B
13 text(results.L_total, 0.15, 'Source B', 'HorizontalAlignment', 'center')
14 ;
15 for i = 1:length(results.loads)
16     x_pos = results.loads(i).distance;
17     plot([x_pos x_pos], [0 -1], 'k-', 'LineWidth', 1); % Load branch
18     plot(x_pos, -1, 'v', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); %
19     Load arrow
20     label_str = sprintf('Load %d\\n%.1f A', i, results.loads(i).current)
21     ;
22     text(x_pos, -1.1, label_str, 'HorizontalAlignment', 'center', '
23     VerticalAlignment', 'top');
24 end
25 title('Unifilar Network Diagram');
26 xlabel('Distance (m)');
27 set(gca, 'YTick', []);
28 ylim([-2 1]);
29 end

```

Listing 6: I4\_PlotUnifilarDiagram.m

### D.3 Module J: Ring and Meshed Network Analysis

```

1 function results = J1_Ring_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Ring Network Analysis
5 % =====
6 results = [];
7 scenarioChoice = menu('Select Ring/Meshed Scenario:', ...
8     'Simple Ring Network (Student A)', ...
9     'Double Ring Network (Conceptual)', ...
10    'Meshed Network (Conceptual)');
11 if scenarioChoice == 0, return; end
12 if scenarioChoice > 1
13     msgbox('Analysis for double-ring and meshed networks requires
14     advanced nodal analysis methods. This module will proceed with the
15     functional Simple Ring analysis as per the coursework.', 'Conceptual
16     Notice', 'warn');
17 end
18 results.scenarioName = 'Simple Ring Network';
19 Ua = input('Enter the feed-in voltage [V]: ');
20 L_total = input('Enter the total circumference of the ring [m]: ');
21 % ... (rest of the input and calculation code)
22 end

```

Listing 7: J1\_Ring\_Analysis.m

```

1 function J2_PlotRingProfile(results)
2 %
3 % =====
4 % FUNCTION: Plot Ring Voltage Profile (as an unwrapped line)
5 % =====
6
7 figure;
8 hold on; box on; grid on;
9 sorted_loads = sortrows(struct2table(results.loads), 'distance');
10 distances = [0, sorted_loads.distance, results.L_total];
11 voltages = [results.Ua];
12 current_in_segment = results.Ia;
13 for i = 1:height(sorted_loads)
14     prev_dist = distances(i);
15     segment_length = sorted_loads.distance(i) - prev_dist;
16     drop = (1/(results.sigma*results.crossSection)) * current_in_segment
17         * segment_length;
18     voltages(end+1) = voltages(end) - drop;
19     current_in_segment = current_in_segment - sorted_loads.current(i);
20 end
21 voltages(end+1) = results.Ua;
22 plot(distances, voltages, '-ob', 'LineWidth', 2, 'MarkerFaceColor', 'r')
23 ;
24 plot(results.min_voltage_node_distance, results.min_voltage, 'kd', '
25     MarkerSize', 10, 'MarkerFaceColor', 'y');
26 title(['Voltage Profile for Ring Network (Unwrapped)']);
27 xlabel('Distance from Feed Point along Ring (m)');
28 ylabel('Voltage (V)');
29 legend('Voltage Profile', 'Minimum Voltage', 'Location', 'southwest');
30 end

```

Listing 8: J2\_PlotRingProfile.m

```

1 function J3_PlotUnifilarDiagram(results)
2 %
3 % =====
4 % FUNCTION: Generates a unifilar schematic diagram of the ring network.
5 % =====
6
7 figure;
8 hold on; axis equal; box on;
9 radius = 1;
10 center_x = 0;
11 center_y = 0;
12 theta = linspace(0, 2*pi, 200);
13 plot(center_x + radius * cos(theta), center_y + radius * sin(theta), 'k-
14     ', 'LineWidth', 4);
15 feed_angle = pi/2; % Place at the top
16 feed_x = center_x + radius * cos(feed_angle);
17 feed_y = center_y + radius * sin(feed_angle);
18 plot(feed_x, feed_y, 's', 'MarkerSize', 20, 'MarkerFaceColor', 'g');
19 text(feed_x, feed_y + 0.15, 'Feed-In', 'HorizontalAlignment', 'center',
20     'FontWeight', 'bold');
21 % ... (rest of the plotting code)

```

18 `end`

Listing 9: J3\_PlotUnifilarDiagram.m

```

1 function J4_PlotLoadingDiagram(results)
2 %
3 % =====
4 % FUNCTION: Generates a network loading diagram for an unwrapped ring.
5 % =====
6
7 figure;
8 hold on; box on; grid on;
9 distances = [0];
10 currents = [results.Ia];
11 current_in_segment = results.Ia;
12 sorted_loads = sortrows(struct2table(results.loads), 'distance');
13 for i = 1:height(sorted_loads)
14     distances(end+1) = sorted_loads.distance(i);
15     currents(end+1) = current_in_segment;
16     current_in_segment = current_in_segment - sorted_loads.current(i);
17     distances(end+1) = sorted_loads.distance(i);
18     currents(end+1) = current_in_segment;
19 end
20 distances(end+1) = results.L_total;
21 currents(end+1) = -results.Ib;
22 plot(distances, currents, '-r', 'LineWidth', 2);
23 line([0, results.L_total], [0, 0], 'Color', 'k', 'LineStyle', '--');
24 title('Ring Network Loading Diagram (Unwrapped)');
25 xlabel('Distance from Feed Point (m)');
26 ylabel('Current (A)');
27 end

```

Listing 10: J4\_PlotLoadingDiagram.m

## D.4 Module J: Network Comparison Tools

```

1 function results = J5_CompareNetworks()
2 %
3 % =====
4 % FUNCTION: Compare Ring and Radial Networks
5 % =====
6
7 results = [];
8 disp('--- Network Comparison Tool ---');
9 U_source = input('Enter the source/feed-in voltage [V]: ');
10 materialChoice = menu('Select Conductor Material:', 'Copper', 'Aluminum',
11     );
12 % ... (rest of the input and calculation code)
13 end

```

Listing 11: J5\_CompareNetworks.m

```

1 function [total_cost, cable_cost, loss_cost] = J6_CalculateNetworkCost(
2     type, L_total, Ia, Ib, loads, sigma, s, cost_per_meter)

```

```

2 %
  =====
3 % FUNCTION: Calculate Network Lifecycle Cost
4 %
  =====

5 LCC_years = 20;
6 hours_per_year = 4000;
7 cost_per_kWh = 0.15;
8 if strcmp(type, 'ring')
9     cable_length = L_total;
10 else % radial
11     cable_length = loads(end).distance;
12 end
13 cable_cost = cable_length * cost_per_meter;
14 % ... (rest of the cost calculation code)
15 end

```

Listing 12: J6\_CalculateNetworkCost.m

```

1 function J7_PlotComparison(results)
2 %
  =====
3 % FUNCTION: Plot Network Comparison Charts
4 %
  =====

5 % --- FIGURE 1: SCHEMATIC DIAGRAMS ---
6 figure('Name', 'Network Schematics Comparison');
7 subplot(1, 2, 1);
8 % ... (Code to plot ring schematic)
9 title('Ring Schematic');
10 subplot(1, 2, 2);
11 % ... (Code to plot radial schematic)
12 title('Radial Schematic');
13 % --- FIGURE 2: LOADING DIAGRAMS ---
14 % ... (Code for loading diagrams)
15 % --- FIGURE 3: PERFORMANCE & ECONOMIC COMPARISON ---
16 % ... (Code for performance and economic charts)
17 end

```

Listing 13: J7\_PlotComparison.m

```

1 %
  =====
2 % SCRIPT for Network Comparison Analysis and Reporting
3 %
  =====

4 clc;
5 results = J5_CompareNetworks();
6 if isempty(results)
7     disp('Analysis cancelled. No report generated.');
```

```

7     return;
9 end
10 fprintf('\n\n--- COMPREHENSIVE REPORT: NETWORK TOPOLOGY COMPARISON ---\n
    ');

```

```

11 % ... (rest of the reporting script)
12 end

```

Listing 14: J\_Run\_Comparison\_And\_Report.m

## D.5 Master Script

```

1 %
  =====
2 % MASTER SCRIPT for Week 3: Network Topology
3 %
  =====
4 clc;
5 clear;
6 close all;
7 analysisChoice = menu('Select which network type to analyze:', ...
8                       'Radial Networks (Problem 1)', ...
9                       'Networks Fed from Both Ends (Problem 2)',
10                      ...
11                      'Ring and Meshed Networks (Problem 3)',
12                      ...
13                      '--- NETWORK COMPARISON TOOL ---');
14 switch analysisChoice
15     case 1
16         run('H_Run_Analysis_And_Report.m');
17     case 2
18         run('I_Run_Analysis_And_Report.m');
19     case 3
20         run('J_Run_Analysis_And_Report.m');
21     case 4
22         run('J_Run_Comparison_And_Report.m');
23 end

```

Listing 15: Week3\_Master.m